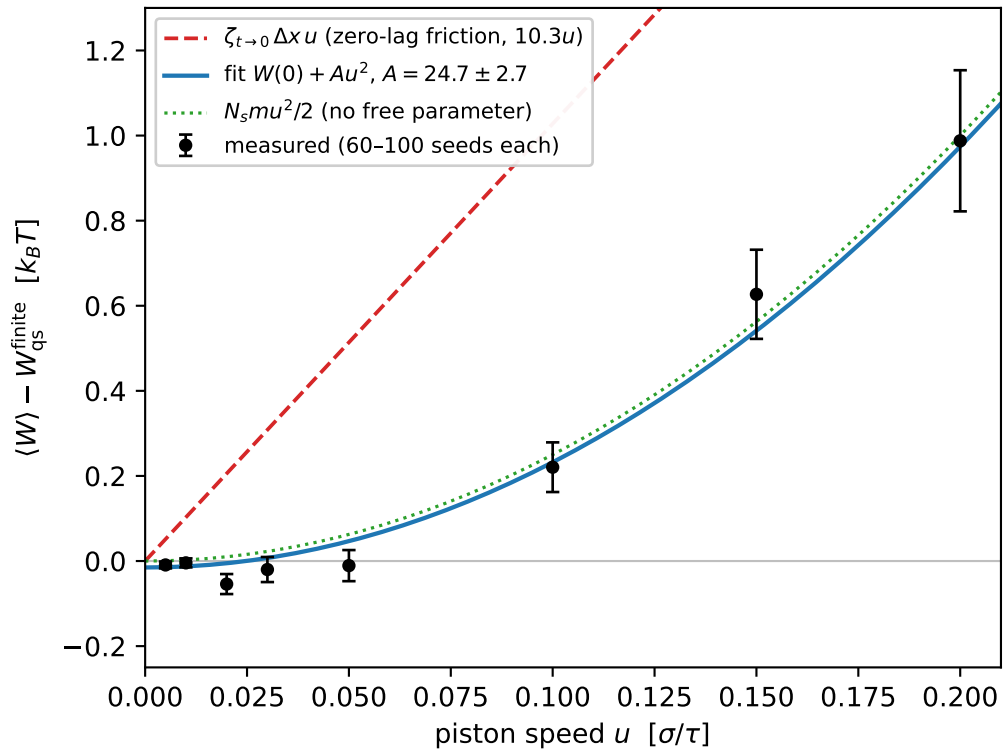


Excess work is second order in the piston speed



The friction integral cancels at the sound traversal time

